Note

Every subtraction statement has a corresponding addition statement. For example, $8-5=3$ is a subtraction statement. This can be seen as the addition statement $3+5=8$ In the same way, $(-8)-(-5)=-3$ is a subtraction statement which can be written as the addition statement $(-8)=(-3)+(-5)$.

Example 1.10 Subtract the following using the number line.

(i) –3 – (–2) (ii) +6 – (–5)

Solution

(i) –3 – (–2)

To subtract –2 from –3 using number line,

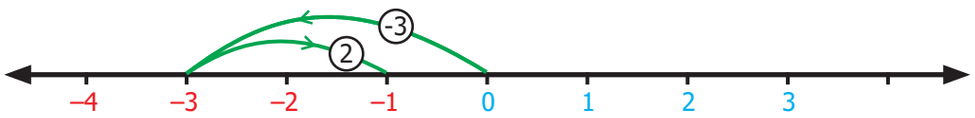


Fig. 1.15

Therefore, –3 – (–2) = –3 + 2 = –1

(ii) +6 – (–5)

To subtract –5 from 6 using number line,

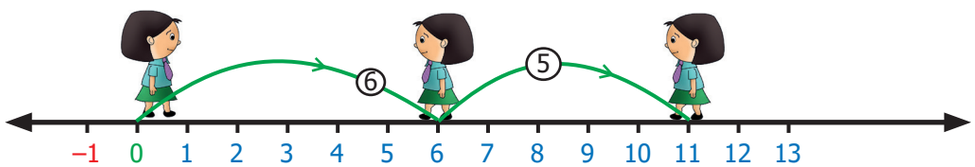


Fig. 1.16

Therefore, +6 – (–5) = +6 + 5 = 11.

Now, let us see how to subtract negative integers using additive inverse.

Example 1.11

(i) Subtract (-40) from 70 (ii) Subtract (-12) from (-20)

Solution

(i) $70-(-40)$

$=70+$ (additive inverse of -40)

$=70+40$

$=110.$

(ii) $(-20)-(-12)$

$=(-20)+$ (additive inverse of (-12))

$=(-20)+12$

$=-8$

Example 1.12

Find the value of : (i) $(-11)-(-33)$ (ii) $(-90)-(-50)$

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